

HARDWARE & OPERATIONAL PROTOCOLS

Pocket Smart AI Companion

Technical deployment manual and hardware specifications for edge inference devices.

CHASSIS MODEL

PX-1 Edge Cortex

CORE ARCHITECTURE

Dual NPU 38 TOPS

MANUAL VERIFIED // ACTIVE SYSTEM

System Architecture

On-Device Subsystem

Edge Models: Sub-3B SLM runs directly in onboard memory.

Zero Latency: Local speech-to-text operates in 32ms frames.

Local Storage: Encrypted vector memory indexed on hardware.

Cloud Relay Subsystem

Deep Reasoning: Complex agent tasks routed via secure relay.

Sync Protocol: Encrypted payload transfer via BLE 5.3 and Wi-Fi.

API Gateways: Native connectors for custom LLM endpoints.

Chassis & Component Layout



Tactile Action Trigger

Precision milled microswitch for voice capture and action triggers.

Dual Beamforming Mics

Omnidirectional acoustic sensors with directional noise isolation.

Magnetic Attachment Mount

Neodymium ring array compatible with standard smartphone shells.

Hardware Data Sheet

Subsystem	Engineering Parameter	Operational Specification
Neural Engine	Dual-Core NPU Accelerators	38.4 TOPS INT8 Performance
System Memory	Unified Low-Power LPDDR5X	12 GB High-Speed Buffer
Secure Storage	Hardware-Encrypted UFS 4.0	128 GB On-Device Vault
Acoustic Array	Dual MEMS Beamforming	130 dB SPL Dynamic Range
Wireless Radios	Dual Band RF Module	BLE 5.3 & Wi-Fi 6E (2.4/5GHz)

Initial Setup & Pairing Sequence

01

Unpack & Inspect

Inspect chassis seal and connect USB-C cable.

02

Engage Power

Hold side button for 4 seconds until blue LED pulses.

03

BLE Handshake

Launch companion application and confirm cryptographic pin.

04

Voice Calibration

Read prompt aloud to generate on-device voice profile.

Physical Controls & Input Sequences

Side Actuator Key

Single Click: Start or stop live meeting transcription.

Double Click: Pin bookmark timestamp with urgent priority tag.

Press & Hold (2s): Activate quick conversational query mode.

Bi-Stable Physical Slider

Upward Position: Phone call and direct screen audio capture.

Downward Position: Ambient 360-degree room conversation mode.

Lock Detent: Prevents pocket activation during transit.

Status LED Optical Diagnostics



Solid Blue

SYSTEM READY

Connected to companion host via Bluetooth.



Glow Amber

AUDIO ACTIVE

Microphones capturing speech in real time.



Pulse Green

SYNC PIPELINE

Offloading encrypted logs to mobile memory.



Rapid Red

ATTENTION

Battery below 10% or hardware vault locked.

Acoustic Isolation & Latency

32ms

On-Device Speech-to-Vector Latency

Hardware DSP isolates human speech while cutting ambient noise by 42 dB.

On-Device Neural Engine Tiers

Acoustic Transcribe

LOCAL TIER 01

Converts spoken dialogue directly into structured text tokens.

Vector Semantic Index

LOCAL TIER 02

Embeds conversations into high-dimensional memory vectors.

Action Synthesizer

LOCAL TIER 03

Extracts follow-up commitments into standard calendar tasks.

Microprocessor & Vision Engine

Optical Object Tagging

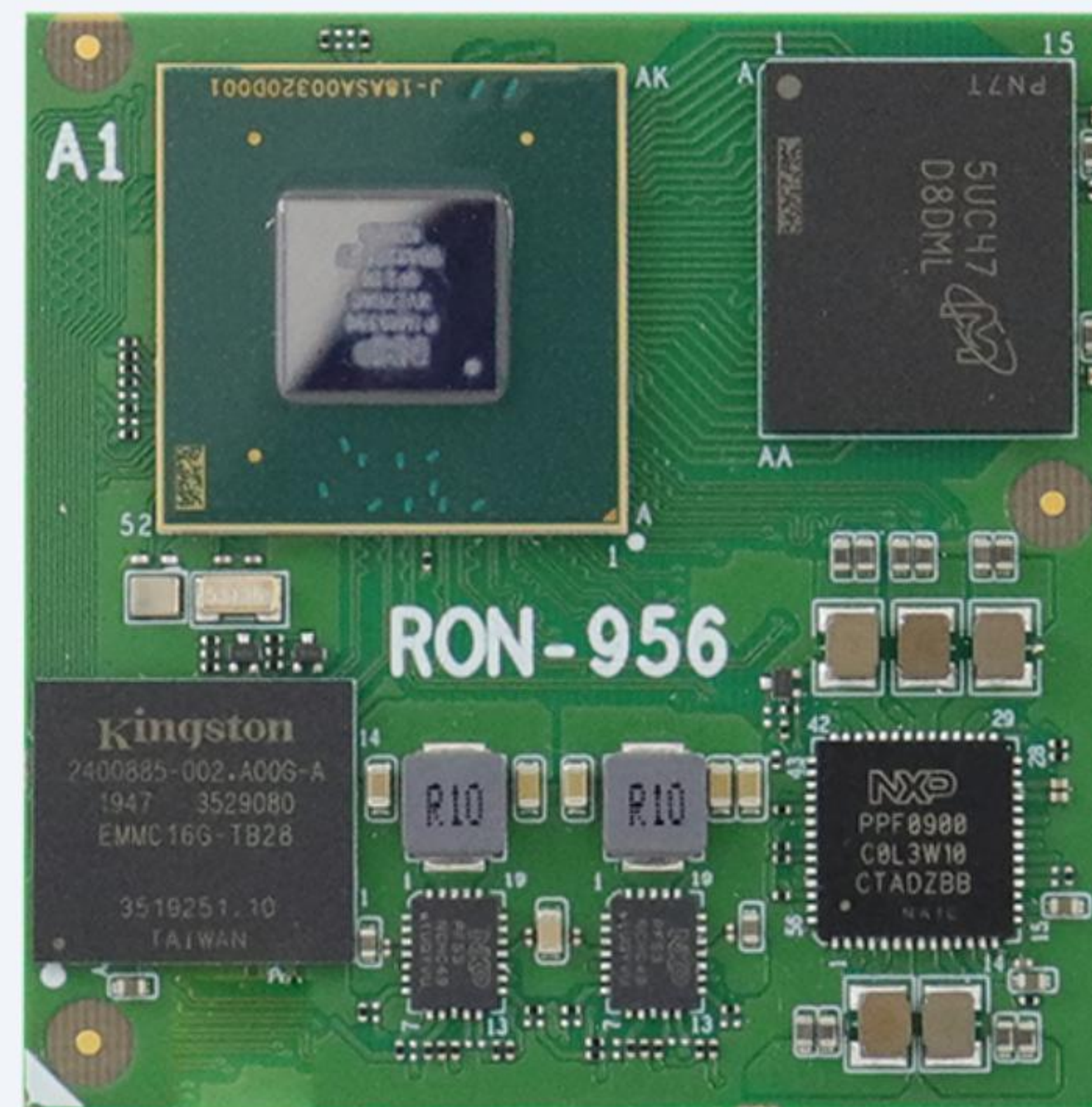
Recognizes documents, whiteboard sketches, and handwritten notes.

Hardware Context Bus

Binds audio streams directly with captured visual frames.

Low-Thermal Throttling

Passive copper heat spreader maintains continuous operation.



Preliminary

Hardware Vault & Privacy Protocols



AES-XTS 256

Hardware encryption keys generated in biometric silicon enclave.



Air-Gap Mode

Disables all radios while processing voice locally.



Speaker Diarization

Identifies approved voices and masks external speech.



Instant Purge

Triple button press permanently wipes memory partitions.

Data Synchronization Protocols

⚡ Wireless BLE 5.3 Channel

Payload: 2 Mbps continuous stream to mobile client.

Range: Up to 15 meters indoor range.

Background Sync: Auto-transfers notes when host connects.

⚡ Wired USB 3.2 Gen 2

Transfer Rate: 10 Gbps desktop web sync pipeline.

Model Reload: Flashes new quantized weights in 18 seconds.

Developer MCP: Exposes local endpoint for tool integration.

Battery Profiles & Discharge Limits



Firmware Updates & Recovery



OTA Download

Signed package transfers to inactive Slot B.



SHA-384 Verify

Cryptographic signature validated in secure enclave.



Atomic Switch

System boots new partition upon next restart.



Safe Rollback

Reverts to Slot A if watchdog triggers boot fault.

Field Troubleshooting Matrix

Symptom	Probable Root Cause	Resolution Procedure
No Response on Trigger	Depleted battery cell or deep sleep	Connect 20W USB-C for 10 minutes
LED Rapid Red Flashing	Storage partition capacity above 98%	Initiate companion app archive sync
Voice Capture Incomplete	Slider set to incorrect mode	Slide downward for ambient room audio
BLE Pairing Timeout	Existing bond conflict on host	Hold trigger 10s to reset radio link



Official Support & Compliance

CE, FCC, and RoHS certified. For replacement parts, developer SDK documentation, or warranty claims, consult your authorized service terminal.

SUPPORT: `help@pocketai.internal`

PORTAL: `docs.pocketai.internal`

Image Sources



<https://preview.redd.it/pi5-portable-ai-v0-uznxi4w86h1g1.jpg?width=640&crop=smart&auto=webp&s=163c5afa901ef466ace17cfe230126e827f49b35>

Source: www.reddit.com



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